EXAM Acidity AQ-70

Explain, in terms of charge/size ratio, why $[Al(H_2O)_6]^{3+}$ is more acidic than $[Fe(H_2O)_6]^{2+}$. [3]

EXAM Amphoteric aluminium AQ-72

Use equations to show that aluminium hydroxide is amphoteric. [3]

EXAM Carbonate AQ-71

Describe and explain what you see when $Na_2CO_3(aq)$ is added separately to $Fe^{2+}(aq)$ and $Fe^{3+}(aq)$. [4]

EXAM Excess ammonia AQ-73

Describe what you see when $NH_3(aq)$ is added dropwise then in excess to $Cu^{2+}(aq)$. Give equations. [4]

EXAM Chloride ions AQ-74

Concentrated HCl is added to $[Cu(H_2O)_6]^{2+}(aq)$. Give the equation, the colour change, the new co-ordination number and shape. [4]

EXAM Hydroxide precipitates AQ-76

$Fe^{2+}(aq)$ is treated with NaOH(aq) and the mixture is left in air. Describe all observations and explain the final change. [3]

EXAM Amphoteric aluminium AQ-72

1. States amphoteric = reacts with acids and bases
2. $[Al(H_2O)_3(OH)_3] + 3H^+ \rightarrow [Al(H_2O)_6]^{3+}$
3. $[Al(H_2O)_3(OH)_3] + OH^- \rightarrow [Al(OH)_4]^- + 3H_2O$

EXAM Acidity AQ-70

1. Al^{3+} is **smaller** and has a **higher charge**, so a higher charge/size ratio (charge density)
2. So it is **more polarising** and **weakens the O–H bonds** in the water ligands
3. So **H^+ is released more easily**

All three links needed for 3 marks.

EXAM Carbonate AQ-71

1. Fe^{2+}: **green ppt** of $FeCO_3$, no effervescence
2. Fe^{3+}: **brown ppt** and **effervescence**
3. $[Fe(H_2O)_6]^{3+}$ is **more acidic** (higher charge/size): releases H^+/H_3O^+
4. H_3O^+ reacts with CO_3^{2-} to give **CO_2**; the 2+ ion is not acidic enough

EXAM Excess ammonia AQ-73

1. **Blue ppt** forms
2. $[Cu(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Cu(H_2O)_4(OH)_2] + 2NH_4^+$
3. Ppt dissolves to a **deep blue solution**
4. $[Cu(H_2O)_4(OH)_2] + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 2OH^- + 2H_2O$

EXAM Hydroxide precipitates AQ-76

1. **Green ppt** forms ($[Fe(H_2O)_4(OH)_2]$)
2. **No change in excess** NaOH
3. On standing it turns **brown**: Fe(II) oxidised to Fe(III) by **oxygen in air**

EXAM Chloride ions AQ-74

1. $[Cu(H_2O)_6]^{2+} + 4Cl^- \rightleftharpoons [CuCl_4]^{2-} + 6H_2O$
2. **Blue → yellow** (green) solution
3. Co-ordination number **4**
4. **Tetrahedral** (Cl^- larger, so fewer fit)

EXAM Chloride ions AQ-75

Compare the substitution of H_2O by NH_3 and by Cl^- in $[Cu(H_2O)_6]^{2+}$, explaining any change in co-ordination number. [4]

EXAM Identify & practical AQ-77

Solution X gives a white ppt with NaOH(aq) that dissolves in excess, and a white ppt with effervescence with Na_2CO_3(aq). Identify X and write an equation for each reaction. [4]

EXAM Identify & practical not in notes AQ-78

Required practical 11. You have four unlabelled solutions: Cu^{2+}, Fe^{2+}, Fe^{3+} and Al^{3+}. Describe how to identify each using test-tube reactions. [6, levels] [6]

EXAM Excess ammonia AQ-79

Describe what you see when NH_3(aq) is added dropwise then in excess to separate solutions of Fe^{2+}, Cu^{2+}, Fe^{3+} and Al^{3+}. Explain the differences. [6, levels] [6]

EXAM Chloride ions AQ-75

1. NH_3: co-ordination number **stays 6**, octahedral
2. NH_3 and H_2O are **similar size** and **uncharged**
3. Cl^-: co-ordination number **6 → 4**, tetrahedral
4. Cl^- is **larger** than H_2O, so fewer fit

The explanation for NH_3 is from the AQA spec (3.2.5).

EXAM Identify & practical AQ-77

1. X contains **Al^{3+}** ($[Al(H_2O)_6]^{3+}$)
2. $[Al(H_2O)_6]^{3+} + 3OH^- \rightarrow [Al(H_2O)_3(OH)_3] + 3H_2O$
3. $[Al(H_2O)_3(OH)_3] + OH^- \rightarrow [Al(OH)_4]^- + 3H_2O$
4. $2[Al(H_2O)_6]^{3+} + 3CO_3^{2-} \rightarrow 2[Al(H_2O)_3(OH)_3] + 3CO_2 + 3H_2O$

EXAM Identify & practical not in notes AQ-78

1. Small sample of each in a test tube; add **NaOH(aq) dropwise**, then **in excess**
2. Cu^{2+} **blue** ppt; Fe^{2+} **green** ppt (browns in air); Fe^{3+} **brown** ppt
3. Al^{3+} **white** ppt that **dissolves in excess** NaOH
4. Confirm with NH_3: only Cu^{2+} ppt dissolves to **deep blue**
5. Or Na_2CO_3: Fe^{3+} and Al^{3+} **effervesce**; Fe^{2+}/Cu^{2+} do not
6. Initial colours also help: blue, green, yellow-brown, colourless

Levels marking: 5–6 marks needs a clear method plus correct results for all four ions, including the excess step.

EXAM Excess ammonia AQ-79

1. Fe^{2+}: green ppt, no change in excess
2. Cu^{2+}: blue ppt, dissolves in excess → **deep blue solution**
3. Fe^{3+}: brown ppt, no change in excess
4. Al^{3+}: white ppt, **no change** in excess
5. Ppt forms because NH_3 acts as a base, removing H^+ until the complex is neutral
6. Only Cu^{2+} undergoes **ligand substitution** with NH_3; NH_3 too weak a base to dissolve $Al(OH)_3$

For full marks give a correct equation too, e.g. $[Fe(H_2O)_6]^{3+} + 3NH_3 \rightarrow [Fe(H_2O)_3(OH)_3] + 3NH_4^+$.

EXAM Complex ions AQ-80

Describe the bonding in $[Fe(H_2O)_6]^{3+}$ and state its shape. [3]

EXAM Identify & practical AQ-81

Copper metal is extracted from $Cu^{2+}(aq)$ using scrap iron. Give the equation and the colours of the initial and final solutions. [3]

EXAM Identify & practical AQ-82

From $[Cu(H_2O)_6]^{2+}(aq)$, identify W (conc. HCl), X (dilute NH_3), Y (excess NH_3) and Z (Na_2CO_3), with appearances. [6]

EXAM Acidity AQ-83

Suggest why $[Fe(H_2O)_6]^{3+}$ and $[Al(H_2O)_6]^{3+}$ react with carbonate to give CO_2 but $[Cu(H_2O)_6]^{2+}$ does not. [2]

EXAM Identify & practical AQ-81

1. $Fe + Cu^{2+} \rightarrow Fe^{2+} + Cu$
2. Initial solution **blue**
3. Final solution **(pale) green**

From the June 2010 question in your notes.

EXAM Complex ions AQ-80

1. Each H_2O **donates a lone pair** from oxygen to Fe^{3+}
2. Forming 6 **co-ordinate (dative covalent) bonds**
3. **Octahedral**, 90°

EXAM Identify & practical AQ-82

1. W = **$[CuCl_4]^{2-}$**, yellow solution
2. X = **$[Cu(H_2O)_4(OH)_2]$**, blue ppt
3. Y = **$[Cu(NH_3)_4(H_2O)_2]^{2+}$**, deep blue solution
4. Z = **$CuCO_3$**, blue-green ppt

Based on June 2010 Q5 in your notes; each part there also needs an equation.

EXAM Acidity AQ-83

1. 3+ ions have a higher charge/size ratio so their solutions contain more **H_3O^+** (more acidic)
2. Enough H_3O^+ to react with CO_3^{2-} to form **CO_2**; 2+ solutions are not acidic enough, so MCO_3 precipitates